

Multivariate ECG Classification

From a MIT-BIH baseline to a PTB-XL shared-loss fusion model

Kappa Group

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12 ECG channels + clinical metadata

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Project goal and roadmap

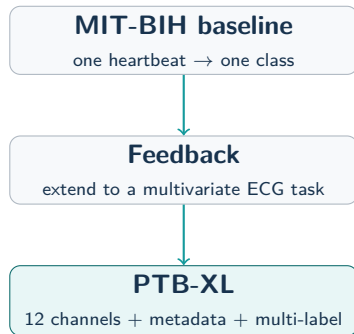
Goal

Build a complete deep learning pipeline for ECG classification.

- ▶ start with a simple heartbeat-level baseline;
- ▶ move to a realistic 12-ECG channels dataset;
- ▶ test whether metadata helps classification.

Main question

Can static clinical metadata improve a 12-ECG channels classifier, and how should it be combined with the ECG signal?



MIT-BIH baseline: validating the workflow

- ▶ Raw ECG records include expert heartbeat annotations.
- ▶ Around each beat, we extract a fixed-size signal window.
- ▶ Each heartbeat belongs to one of five classes.
- ▶ This first task is **multi-class**: one input, one label.

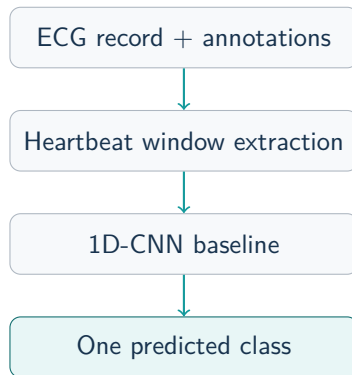
Normal

Supraventricular

Ventricular

Fusion

Unclassifiable



Why move to PTB-XL?

MIT-BIH

- ▶ isolated heartbeat windows;
- ▶ simpler one-label classification;
- ▶ useful baseline, but limited context.

PTB-XL

- ▶ full 10-second ECG recordings;
- ▶ 12 ECG channels recorded together;
- ▶ metadata and diagnostic superclasses;
- ▶ more realistic clinical setting.



Multi-label formulation

MIT-BIH

one heartbeat
one class

- ▶ mutually exclusive classes;
- ▶ one label per example;
- ▶ Cross Entropy Loss.

PTB-XL

one ECG recording
several possible labels

- ▶ output vector: [NORM, MI, STTC, CD, HYP];
- ▶ one logit per diagnostic group;
- ▶ sigmoid probability + threshold per class;
- ▶ Binary Cross Entropy loss during training.

Example after sigmoid

NORM	MI	STTC	CD	HYP
0.18	0.81	0.74	0.22	0.61

Class-specific thresholds: NORM 0.65 | MI 0.70 | STTC 0.50 | CD 0.65 | HYP 0.60

Metadata selection and label leakage

Selected clinical metadata

age

sex

height

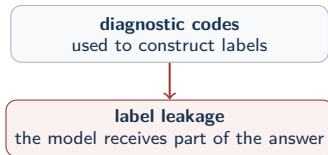
weight

- ▶ Simple patient-level information.
- ▶ Clinical context without diagnostic labels.

Type of information

Static metadata, not temporal ECG signals.

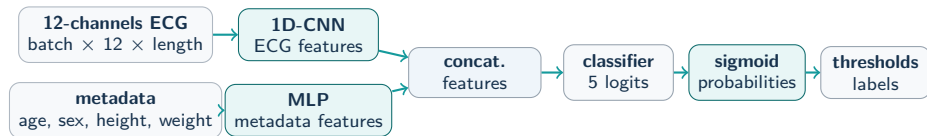
Main risk: label leakage



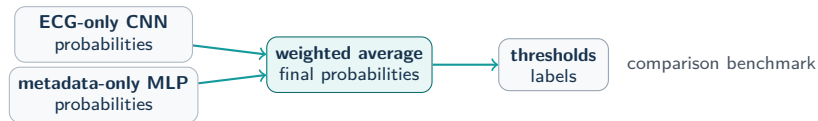
Diagnostic codes were excluded because they are directly related to the target labels.

Fusion architecture

Proposed model: feature-level fusion



Benchmark: soft voting



- ▶ Main model: ECG and metadata are fused before the final classifier.
- ▶ Output path: 5 logits → sigmoid probabilities → thresholds → final labels.
- ▶ Soft voting is shown only as a separate comparison benchmark.

Why CNN for ECG and MLP for metadata?

ECG signal



Temporal signal

- ▶ waveform evolves over time;
- ▶ 1D-CNN learns local temporal features.

Clinical metadata

age

sex

height

weight

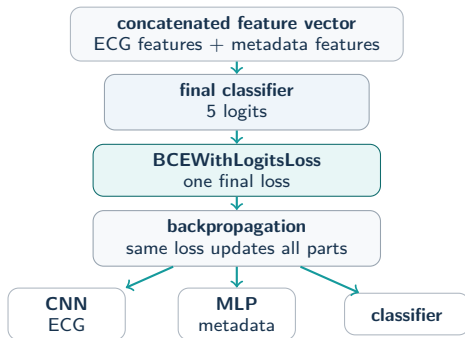
Static tabular data

- ▶ not meaningful as temporal channels;
- ▶ MLP is appropriate for compact tabular input.

Shared loss and backpropagation

End-to-end training

- ▶ After concatenation, one final classifier outputs 5 logits.
- ▶ BCEWithLogitsLoss is computed from these logits.
- ▶ The loss is averaged over labels and batch samples.
- ▶ Backpropagation updates the CNN, MLP and classifier together.



Loss function

$$\mathcal{L} = -\frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C \left[p_c y_{i,c} \log(\sigma(z_{i,c})) + (1 - y_{i,c}) \log(1 - \sigma(z_{i,c})) \right]$$

Training setup

Folds 1–8
Training
17,418 samples

Fold 9
Validation
2,183 samples

Fold 10
Test
2,198 samples

Training parameters

- ▶ Adam optimizer
- ▶ Learning rate: 0.001
- ▶ Weight decay: 10^{-4}
- ▶ BCEWithLogitsLoss
- ▶ Positive class weights

Model selection

- ▶ Early stopping on validation macro-F1
- ▶ Best checkpoint kept during training
- ▶ Decision thresholds tuned on validation set

Macro-F1

Training class ratios

NORM	MI	STTC	CD	HYP
43.61%	25.14%	24.03%	22.43%	12.17%

Validation and test ratios were comparable.

$$F1_c = \frac{2P_c R_c}{P_c + R_c} \quad \text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c$$

Same weight for each diagnostic class.

Soft voting benchmark

Prediction-level fusion

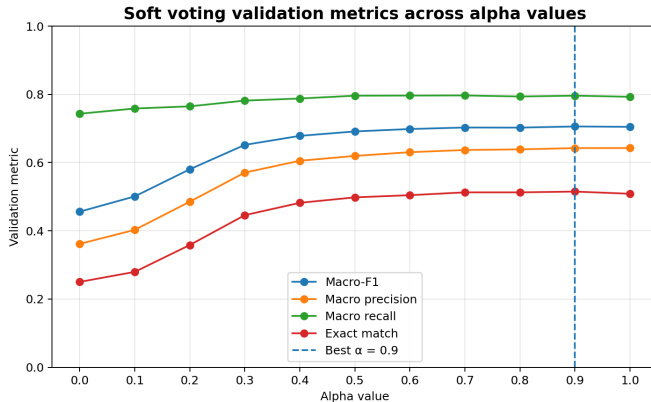
$$p_{\text{soft}} = \alpha p_{\text{ECG}} + (1 - \alpha) p_{\text{metadata}}$$

- ▶ ECG-only and metadata-only are trained separately.
- ▶ Their probabilities are combined after prediction.
- ▶ α controls the ECG contribution.

Validation choice

Best value: $\alpha = 0.9$

Alpha sweep on validation set



MIT-BIH baseline results

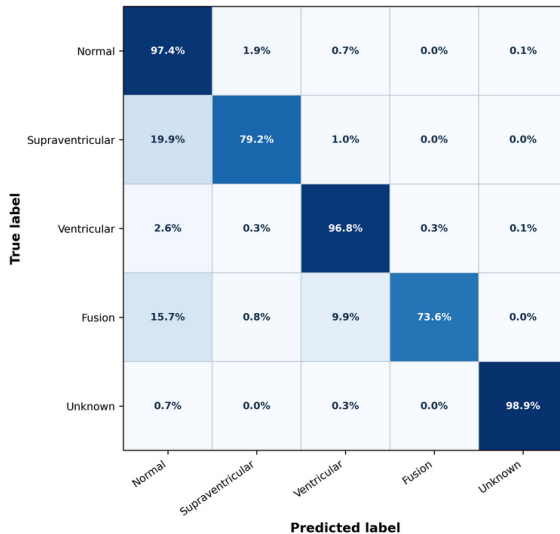
0.971

accuracy

0.864

macro-F1

- ▶ not directly comparable with PTB-XL because the task is simpler.

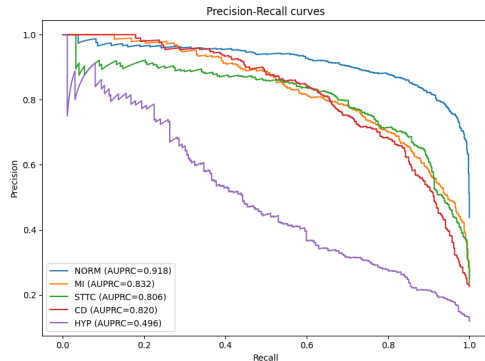
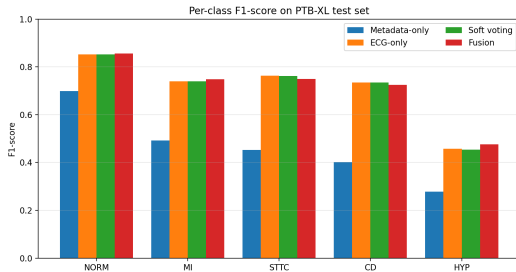


PTB-XL model comparison

Model	Macro-F1	Micro-F1
Metadata-only MLP	0.465	0.491
ECG-only CNN	0.709	0.750
Soft-voting benchmark	0.708	0.749
Shared-loss fusion	0.711	0.754

- ▶ Metadata alone is clearly insufficient.
- ▶ ECG-only is already strong.
- ▶ Shared-loss fusion is best, but the improvement is modest.

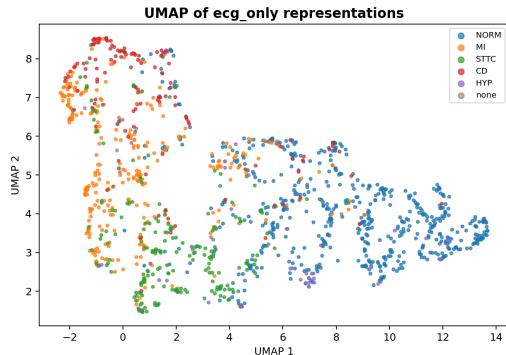
Per-class F1 and precision-recall curves



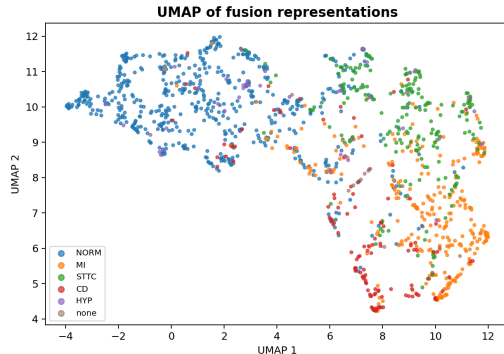
- ▶ HYP is the weakest diagnostic superclass.
- ▶ PR curves are shown for the final CNN + MLP fusion model.
- ▶ Performance is not uniform across diagnostic groups.

Representation check with UMAP

ECG-only model



Fusion model



- ▶ Fusion does not create a radically different separation.
- ▶ This is consistent with the modest gain over ECG-only.

Conclusion and future work

Main conclusions

- ▶ Modular ECG pipeline: MIT-BIH → PTB-XL.
- ▶ Shared-loss fusion gives the best result.
- ▶ Gain over ECG-only is modest.
- ▶ ECG remains the dominant source.

Future work

- ▶ Better imbalance handling.
- ▶ Stronger ECG architectures.
- ▶ Revisit soft voting with calibrated probabilities.